

CENG381 Stochastic Processes

Stationary Distribution - Question Set 1

Q1 (30 points). Three out of every five buses on a route are followed by a car, while only one out of every four cars is followed by a bus. Assume vehicles on the road form a Markov chain with states {Bus, Car}.

- Set up the transition matrix P for this Markov chain.
 - Find the stationary distribution $\pi = (\pi_B, \pi_C)$ by solving $\pi * P = \pi$ and $\pi_B + \pi_C = 1$.
 - Interpret π_B : what fraction of vehicles on the road are buses?
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Q2 (35 points). A frog sits on two lily pads labelled E (east) and W (west). The transition matrix is:

$$\begin{array}{cc} & \begin{array}{cc} E & W \end{array} \\ \begin{array}{c} E \\ W \end{array} & \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix} \end{array}$$

where $p, q > 0$ and $p + q < 1$.

- Verify that $\pi(E) = q/(p+q)$ and $\pi(W) = p/(p+q)$ is a stationary distribution, i.e. show $\pi * P = \pi$.
 - Let M_t be the distribution at time t and $M_0 = (1, 0)$ (starting at E). Write a recurrence for M_t and confirm that $M_t \rightarrow \pi$ as $t \rightarrow \infty$.
 - For $p = 0.3$ and $q = 0.4$, compute π and find the exact value of $P^{(2)}$ using the formula $P^{(2)} = S^{-1} D^2 S$ where $\lambda_1=1$ and $\lambda_2=1-p-q$ are the eigenvalues of P .
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Q3 (35 points). Consider the Markov chain with 3 states {0, 1, 2} and transition matrix:

$$\begin{array}{cc} & \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \end{array} & \begin{bmatrix} 1/2 & 1/3 & 1/6 \\ 1/4 & 1/2 & 1/4 \\ 1/6 & 1/3 & 1/2 \end{bmatrix} \end{array}$$

- Find the stationary distribution $\pi = (\pi_0, \pi_1, \pi_2)$ by solving $\pi * P = \pi$ with $\pi_0 + \pi_1 + \pi_2 = 1$.
- Verify your answer by checking $\pi * P = \pi$ directly.
- Is this chain time-reversible? Check the detailed balance condition $\pi_i * P_{ij} = \pi_j * P_{ji}$ for all pairs (i,j) . What does this tell you about the long-run flow between states?